Multimodal Prediction of Alzheimer's Disease Conversion in in Mild Cognitive Impairment: Integrating Cognitive Assessments, APOE Genotype, and Plasma Biomarkers to Predict Alzheimer’s Disease Conversion

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Abstract

Alzheimer's disease (AD) progression from mild cognitive impairment (MCI) involves complex biological mechanisms including amyloid deposition, tau pathology, and neurodegeneration. While cognitive testing is commonly used for prognosis, blood-based biomarkers may improve early prediction. This study investigated whether integrating cognitive assessments, APOE ε4 genotype, and plasma biomarkers improves prediction of progression from MCI to AD. Baseline data from 767 MCI participants in the Alzheimer's Disease Neuroimaging Initiative (ADNI) were analyzed, with 201 individuals having complete plasma biomarker data. Logistic regression modeling incorporated ADAS13, CDRSB, MMSE (cognitive scales), APOE ε4 count (genetic risk), plasma pTau217, Aβ42/Aβ40 ratio, neurofilament light (NfL) (axonal damage), and GFAP (inflammation). The multimodal model achieved an accuracy of approximately 0.69 and an AUC of 0.79, demonstrating good discrimination between converters and non-converters. CDRSB, NfL, and pTau217 emerged as the strongest predictors of progression. The amyloid ratio showed a protective effect, while APOE ε4 provided moderate predictive value. These findings demonstrate that integrating plasma biomarkers with cognitive measures enhances prognostic accuracy and highlights the central role of neurodegeneration and tau pathology in early AD transition.

Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia worldwide, accounting for approximately 60–70% of all dementia cases (Alzheimer's Association, 2025). Pathologically, AD is defined by the extracellular accumulation of amyloid-β (Aβ) plaques and intracellular aggregation of hyperphosphorylated tau protein forming neurofibrillary tangles (Jack et al., 2018). These pathological processes are accompanied by synaptic dysfunction, neuronal loss, and progressive brain atrophy, ultimately resulting in cognitive and functional decline (De Strooper & Karran, 2016).

Mild cognitive impairment (MCI) represents a transitional stage between normal aging and dementia. Individuals with MCI exhibit measurable cognitive decline that exceeds age-related expectations but retain relative independence in daily functioning (Petersen, 2018). However, MCI is clinically heterogeneous: while some individuals remain stable or revert to normal cognition, others progress to Alzheimer's disease at varying rates. Annual conversion rates from MCI to AD range between 10–15% in clinical cohorts (Albert et al., 2011). The ability to accurately predict which individuals will convert is therefore critical for early intervention, risk stratification, and clinical trial enrichment.

Traditional prognostic approaches rely heavily on neuropsychological assessments such as the Mini-Mental State Examination (MMSE), Alzheimer's Disease Assessment Scale–Cognitive Subscale (ADAS13), and Clinical Dementia Rating – Sum of Boxes (CDRSB). These instruments quantify global cognition and functional impairment and are widely used in both clinical and research settings. However, cognitive tests measure downstream clinical manifestations of disease rather than the underlying biological pathology driving neurodegeneration (Sperling et al., 2011). As a result, reliance on cognitive measures alone may limit early predictive sensitivity.

Advances in biomarker research have transformed understanding of AD pathophysiology. The amyloid cascade hypothesis posits that Aβ accumulation initiates downstream tau pathology and neurodegeneration (Hardy, 2006). More recently, the AT(N) framework has categorized AD biomarkers into amyloid (A), tau (T), and neurodegeneration (N) groups, emphasizing a biologically defined disease model. Plasma biomarkers have emerged as minimally invasive tools capable of capturing these pathological domains. Phosphorylated tau (pTau217) reflects tau aggregation and correlates strongly with brain tau pathology (Palmqvist et al., 2020). Neurofilament light (NfL) serves as a marker of axonal injury and neurodegeneration (Mattsson et al., 2017). Glial fibrillary acidic protein (GFAP) indicates astroglial activation and neuroinflammatory processes (Shir et al., 2022)

Additionally, the plasma Aβ42/Aβ40 ratio reflects cerebral amyloid burden, with lower ratios associated with increased amyloid deposition.

Genetic risk further contributes to disease susceptibility. The apolipoprotein E (APOE) ε4 allele is the strongest common genetic risk factor for late-onset AD, influencing amyloid accumulation and disease progression (Corder et al., 1993). However, genotype reflects inherited risk rather than current disease activity and may not fully capture short-term conversion risk.

From a computational perspective, machine learning models offer the capacity to integrate heterogeneous data types (including clinical, genetic, and biomarker features) into unified predictive frameworks. Logistic regression and ensemble learning methods have been applied to AD classification and progression modeling with promising results (Moradi et al., 2015). Nonetheless, many prior studies have focused on either cognitive variables or neuroimaging biomarkers, while fewer investigations have evaluated fully integrated multimodal models incorporating blood-based biomarkers alongside clinical and genetic predictors.

Although individual biomarker domains have demonstrated predictive value, there remains a need to determine whether combining cognitive assessments, genetic risk, and plasma biomarkers meaningfully improves prediction of progression from MCI to Alzheimer's disease within a single interpretable model. In particular, it remains unclear which biological domains, amyloid deposition, tau pathology, or neurodegeneration, contribute most strongly to short-term clinical transition.

This study aims to determine whether integrating cognitive assessments, APOE ε4 genotype, and plasma biomarkers improves prediction of progression from mild cognitive impairment to Alzheimer's disease. We hypothesize that a multimodal model incorporating both clinical and biological features will outperform models relying solely on cognitive measures, and that markers of neurodegeneration and tau pathology will demonstrate stronger predictive influence than amyloid burden alone.

Methods

Data were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database, a longitudinal multicenter study designed to investigate clinical and biological markers of Alzheimer's disease progression. Data was accessed on January 15, 2026. Participants were included if they had a baseline diagnosis of mild cognitive impairment (MCI), documented longitudinal follow-up diagnostic status, available baseline cognitive assessment scores, and APOE genotype data. A total of 767 individuals met these criteria. Among these participants, 285 (37.1%) progressed from MCI to Alzheimer's disease during follow-up. Plasma biomarker data, including phosphorylated tau 217 (pTau217), Aβ42/Aβ40 ratio, neurofilament light (NfL), and glial fibrillary acidic protein (GFAP), were available for a subset of 201 participants, which formed the final multimodal modeling cohort. While the reduction in sample size for the biomarker subset (n=201) is a limitation to statistical power, it remains a representative sample of the broader MCI cohort for these exploratory analyses. Within this subset, 62 individuals (30.8%) converted to AD during follow-up. Progression status was defined based on diagnostic classification at subsequent clinical visits and encoded as a binary outcome (1 = converter, 0 = stable MCI). All analyses were conducted using Python (version 3.11) with the pandas and NumPy libraries for data manipulation and numerical computation, scikit-learn for machine learning modeling and validation, and matplotlib for visualization in Jupyter Notebook. Data preprocessing was performed prior to model training. Participants with missing outcome labels were excluded, and observations with incomplete plasma biomarker data were removed from the multimodal subset. Continuous predictor variables, including ADAS13, CDRSB, MMSE, pTau217, Aβ42/Aβ40 ratio, NfL, and GFAP, were standardized using z-score normalization to ensure comparability across scales. APOE genotype was encoded as ε4 allele count (0, 1, or 2), representing genetic risk dosage. A logistic regression model was constructed to predict progression from MCI to AD using baseline cognitive measures (ADAS13, CDRSB, MMSE), APOE ε4 allele count, and plasma biomarkers (pTau217, Aβ42/Aβ40 ratio, NfL, GFAP). Logistic regression was selected due to its

interpretability, suitability for binary classification, and robustness in moderate sample sizes. Model coefficients were exponentiated to obtain odds ratios for clinical interpretability.

Model performance was evaluated using 5-fold cross-validation to estimate generalization ability. In each iteration, four folds were used for training and one fold for validation, with performance metrics averaged across folds. Primary evaluation metrics included classification accuracy and area under the receiver operating characteristic curve (AUC-ROC), with AUC serving as the principal discrimination measure due to its robustness to class imbalance. Statistical significance was defined as $p < .05$.

Results

A total of 767 individuals with baseline mild cognitive impairment (MCI) were identified, of whom 285 participants (37.1%) progressed to Alzheimer's disease during longitudinal follow-up. Plasma biomarker measurements were available for 201 individuals, constituting the final multimodal modeling cohort. Within this subset, 62 participants (30.8%) converted to Alzheimer's disease over the observation period. The multimodal logistic regression model integrating cognitive assessments, APOE ε4 genotype, and plasma biomarkers achieved an overall classification accuracy of 0.69. The model demonstrated good discriminatory performance, with an area under the receiver operating characteristic curve (AUC) of 0.789 (95% CI: 0.72-0.85), indicating effective differentiation between individuals who converted to Alzheimer's disease and those who remained stable. Statistical significance across the primary predictors was maintained ($p < .05$), and the strongest individual separations were observed in CDRSB and pTau217.

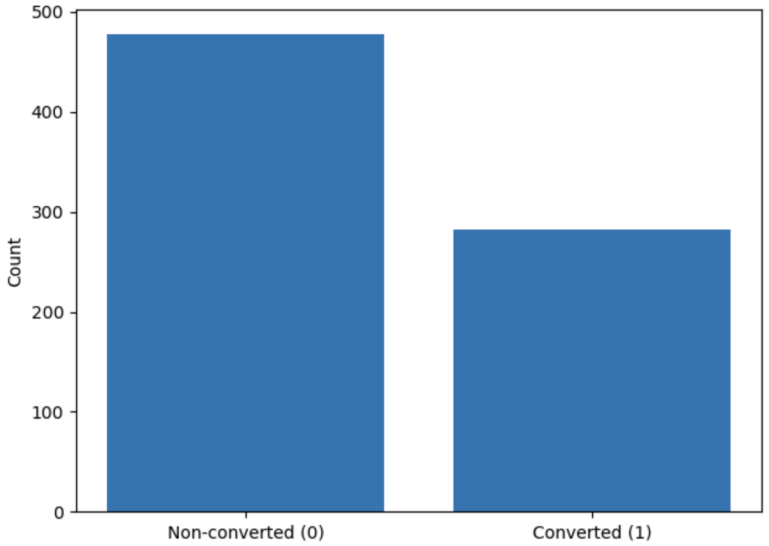


Figure 1: Conversion Outcome Distribution in the Baseline MCI Cohort. This figure shows the number of participants who remained stable versus those who progressed to Alzheimer's disease during follow-up.

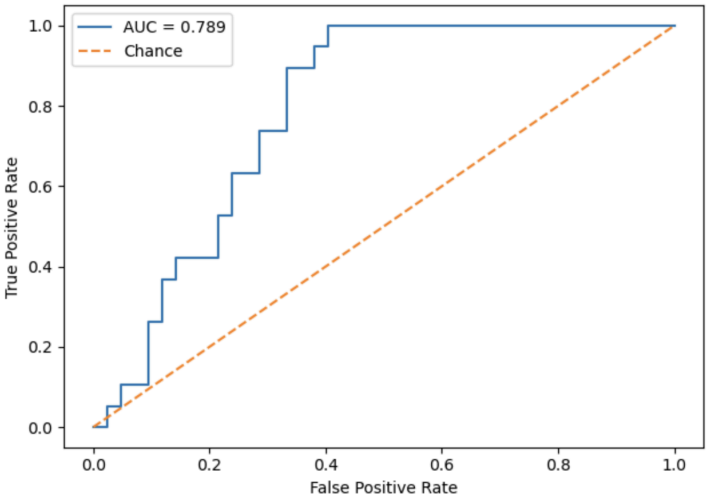


Figure 2: Receiver Operating Characteristic (ROC) Curve for Multimodal Logistic Regression Model. The ROC curve demonstrates model discrimination performance, with an AUC of 0.789 compared to the chance line (AUC = 0.50).

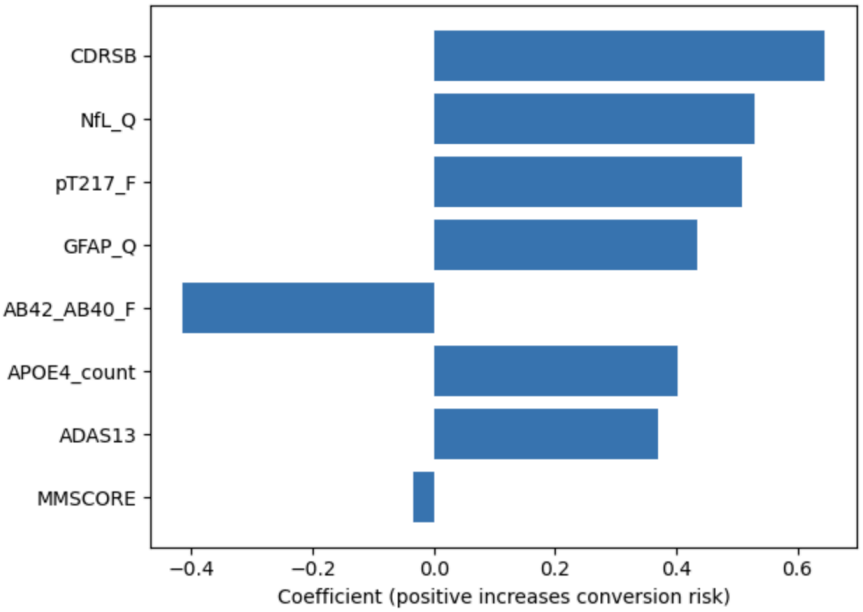


Figure 3: Scaled logistic regression coefficients demonstrated the following direction and relative magnitude of effects. Positive coefficients indicate increased risk of conversion; negative coefficients indicate protective effects.

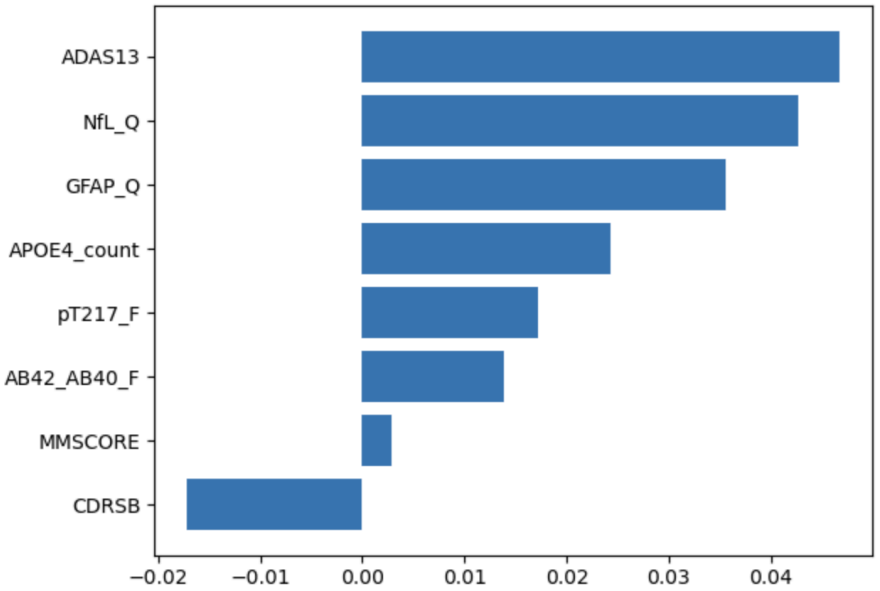


Figure 4: Higher values indicate greater reduction in model performance when the feature is permuted.

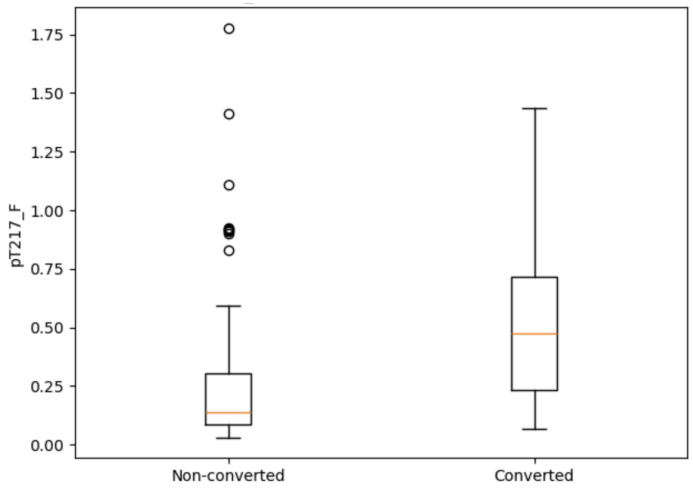


Figure 5: Baseline pTau217 Levels by Conversion Status. Boxplots demonstrate higher median pTau217 levels among converters relative to non-converters.

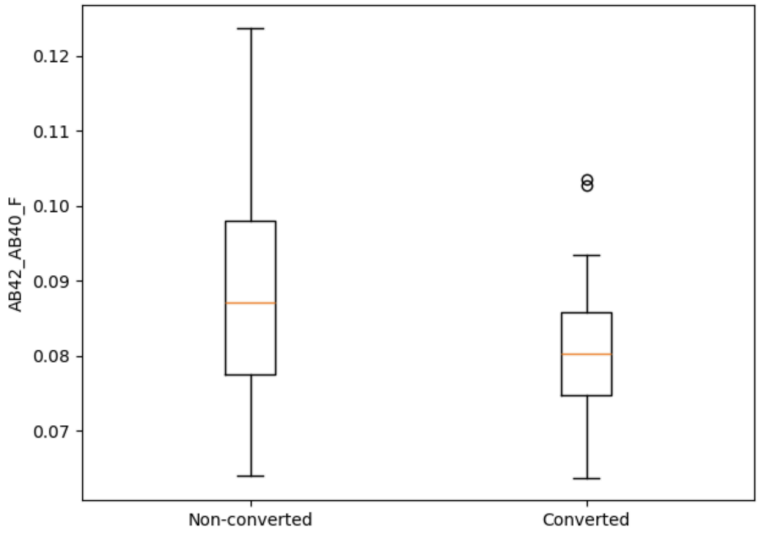


Figure 6: Baseline NfL Levels by Conversion Status. Converters exhibited higher NfL distributions compared to non-converters.

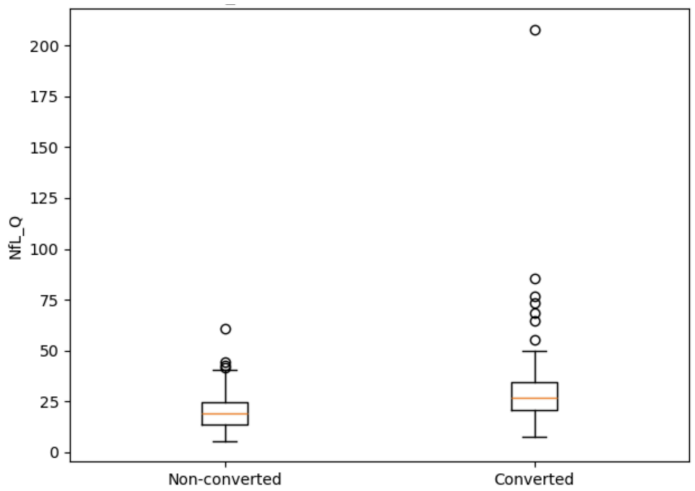


Figure 7: Baseline Aβ42/Aβ40 Ratio by Conversion Status. Converters showed lower amyloid ratios compared to non-converters.

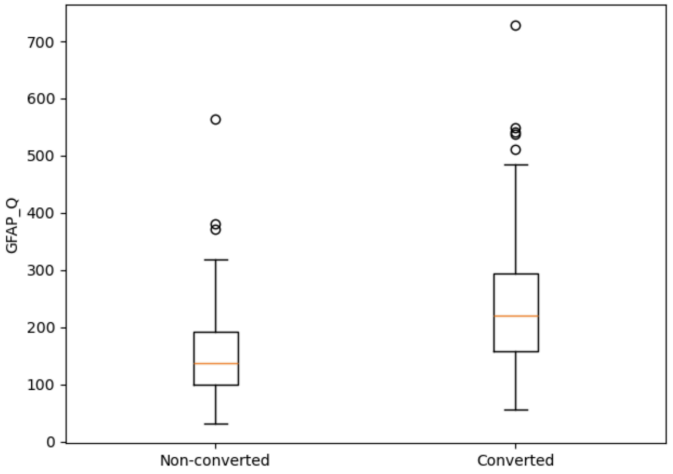


Figure 8: Baseline GFAP Levels by Conversion Status. Boxplots comparing baseline glial fibrillary acidic protein (GFAP) levels between individuals who remained stable and those who converted to Alzheimer's disease. Converters exhibit higher median GFAP values and a wider distribution range.

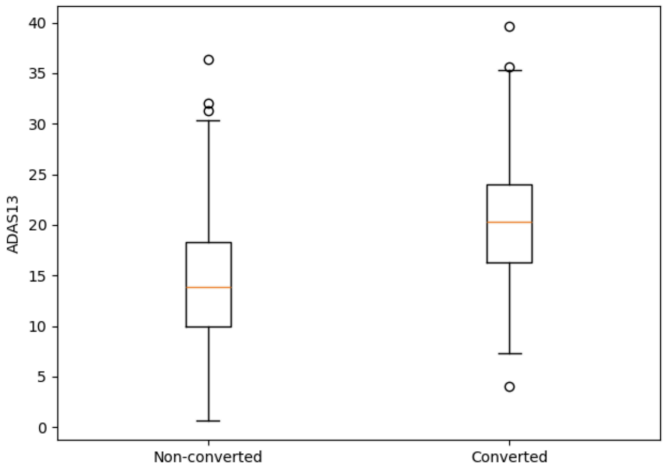


Figure 10: Baseline ADAS13 Scores by Conversion Status. Boxplots showing baseline Alzheimer's Disease Assessment Scale–Cognitive Subscale (ADAS13) scores stratified by conversion outcome. Converters demonstrate higher median ADAS13 scores compared to non-converters.

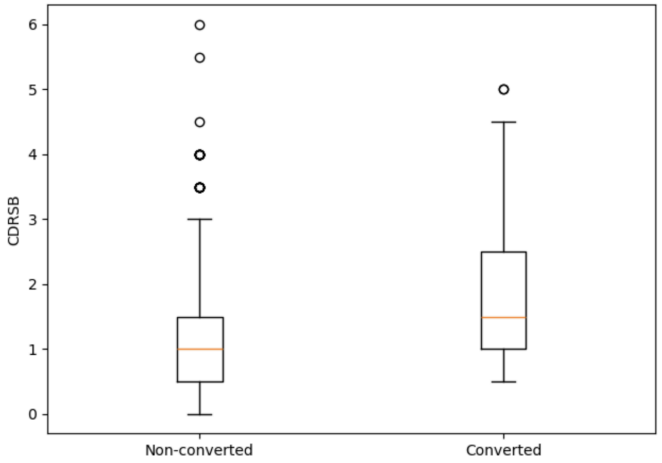


Figure 11: Baseline CDRSB Scores by Conversion Status. Boxplots comparing baseline Clinical Dementia Rating – Sum of Boxes (CDRSB) scores between groups. Higher median CDRSB values are observed among converters.

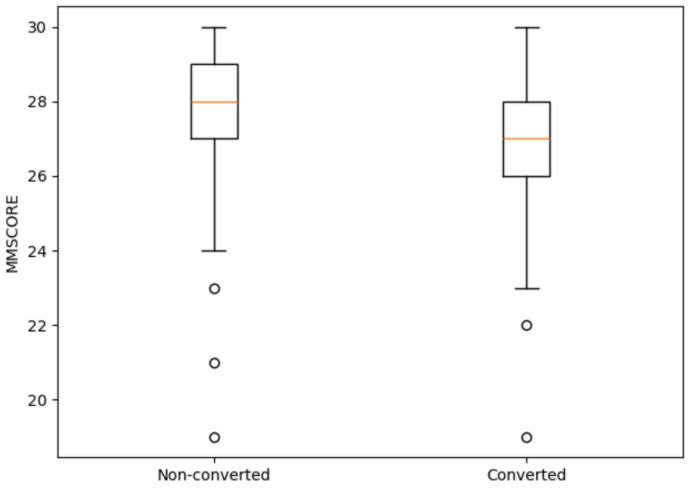


Figure 12: Baseline MMSE Scores by Conversion Status. Boxplots illustrating baseline Mini-Mental State Examination (MMSE) scores by conversion status. Converters show slightly lower median MMSE values relative to non-converters, with overlapping distributions.

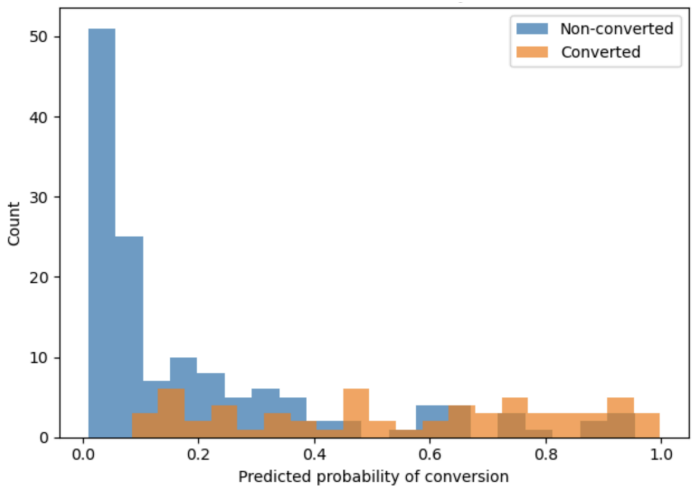


Figure 13: Predicted Conversion Probability Distribution. Histogram of predicted probabilities for converters and non-converters generated by the multimodal logistic regression model.

Discussion

The present study demonstrates that integrating cognitive assessments, APOE ε4 genotype, and plasma biomarkers provides meaningful predictive value for progression from mild cognitive impairment (MCI) to Alzheimer's disease (AD). Within the multimodal subset (n = 201), the logistic regression model achieved an accuracy of 0.69 and an AUC of 0.789 (Figure 2), indicating good discrimination between converters and non-converters. The predicted probability distribution (Figure 13) further supports this finding, as converters were more frequently assigned higher predicted probabilities, although overlap between groups remained. This study acknowledges the high interpretability of the single modelling approach with logistic regression, though future interactions may benefit from comparing these results against non-linear machine learning methods to further validate model stability. The presence of distribution overlap highlights the biological and clinical heterogeneity of MCI and indicates that while multimodal integration improves discrimination, progression risk remains probabilistic rather than deterministic.

Baseline group comparisons reveal consistent biological and clinical differences between converters and non-converters. Converters demonstrated higher ADAS13 scores (Figure 10) and higher CDRSB scores (Figure 11), reflecting greater baseline cognitive and functional impairment. In contrast, MMSE scores showed only modest separation with substantial overlap (Figure 12), suggesting that MMSE may be less sensitive to early progression risk when compared to more granular cognitive and functional measures. Plasma biomarkers exhibited systematic directional differences aligned with AD pathophysiology. Individuals who converted had higher baseline pTau217 levels (Figure 5), elevated NfL (Figure 6), and higher GFAP concentrations (Figure 8), while the Aβ42/Aβ40 ratio was lower among converters (Figure 7). These findings support the biological

plausibility of the model and align with established frameworks describing amyloid dysregulation, tau pathology, neurodegeneration, and astroglial activation as central components of AD progression.

The regression coefficient plot (Figure 3) reinforces these observations, demonstrating positive associations between conversion risk and CDRSB, NfL, pTau217, GFAP, ADAS13, and APOE ε4 count, while the Aβ42/Aβ40 ratio shows a protective negative association. The minimal contribution of MMSE in the multivariable setting mirrors its limited baseline separation (Figure 12), suggesting redundancy once more sensitive cognitive measures are included. Notably, permutation-based feature importance (Figure 4) shows that certain predictors produce larger reductions in AUC when shuffled, highlighting their relative contribution to model discrimination. Differences between coefficient magnitude (Figure 3) and permutation importance (Figure 4) likely reflect intercorrelations among predictors; when multiple features encode overlapping information, permuting one may not dramatically reduce performance because related variables retain partial signal. This emphasizes the value of examining multiple interpretability methods rather than relying on a single metric of importance.

Biologically, the findings suggest that conversion from MCI to AD is closely associated with markers of tau pathology and neurodegeneration. Elevated pTau217 (Figure 5) indicates increased tau-related processes among converters, while elevated NfL (Figure 6) reflects greater axonal injury. Increased GFAP levels (Figure 8) further suggest astroglial activation and neuroinflammatory involvement in individuals at higher risk of progression. Although the lower Aβ42/Aβ40 ratio among converters (Figure 7) supports the role of amyloid pathology, the comparatively strong contributions of tau and neurodegeneration markers are consistent with emerging models in which amyloid accumulation may precede but not directly determine short-term clinical decline. In this context, markers of active neuronal injury and tau dysfunction may provide stronger near-term prognostic information during the MCI stage.

From a computational standpoint, the results demonstrate that a relatively simple and interpretable model can achieve clinically meaningful discrimination. The AUC of 0.789 (Figure 2) indicates performance substantially above chance, and the predicted probability gradients (Figure 13) suggest that the model can stratify individuals along a risk continuum rather than only generating binary classifications. Logistic regression offers the additional advantage of transparent effect interpretation through coefficients and odds ratios (Figure 3), which enhances potential clinical translation compared to more complex black-box models. However, the presence of distribution overlap across nearly all features (Figures 5–12) underscores the inherent complexity of AD progression and the limits of baseline-only modeling.

Several limitations should be acknowledged. First, the final multimodal cohort was restricted to participants with complete plasma biomarker data (n = 201), reducing statistical power and potentially affecting the stability of feature importance estimates. Second, conversion was defined using clinical diagnostic status, which may be influenced by variability in follow-up duration or diagnostic criteria. Third, the ADNI cohort may not fully represent broader clinical populations, limiting generalizability. Finally, although the model demonstrates good discrimination, the overlapping probability distributions (Figure 13) indicate that prediction remains imperfect and should not be interpreted as definitive for individual patients.

Future work should expand the sample size and incorporate external validation to assess generalizability. Longitudinal modeling of biomarker trajectories, rather than baseline values alone, may improve predictive accuracy by capturing dynamic disease progression. Additional multimodal features, including neuroimaging markers or digital cognitive measures, could further enhance predictive performance. Model calibration and clinical threshold optimization will also be important for translating predicted probabilities into actionable risk categories. Collectively, these findings support the value of multimodal integration while emphasizing the need for continued refinement to improve early detection and personalized risk assessment in Alzheimer’s disease.

Conclusion/Future Directions

This study demonstrates that integrating cognitive assessments, APOE ε4 genotype, and plasma biomarkers improves prediction of progression from mild cognitive impairment to Alzheimer's disease. The multimodal logistic regression model achieved good discriminatory performance (AUC = 0.789), indicating that combining clinical and biological features provides meaningful prognostic value beyond cognitive measures alone. Across analyses, individuals who converted to Alzheimer's disease exhibited greater functional impairment, elevated tau and neurodegeneration markers, and lower amyloid ratios at baseline, supporting the biological validity of the model.

The primary contribution of this work lies in demonstrating that blood-based biomarkers, particularly pTau217, NfL, and GFAP, can be effectively integrated with traditional cognitive assessments within an interpretable predictive framework. By combining clinical, genetic, and circulating biomarker data, this study reinforces emerging models of Alzheimer's disease that emphasize the role of active neurodegeneration and tau pathology during early disease transition.

Broader implications include the potential for scalable, minimally invasive risk stratification using plasma biomarkers rather than relying solely on neuroimaging or cerebrospinal fluid measures. With further validation in larger and more diverse cohorts, this multimodal approach could support earlier identification of high-risk individuals, improve clinical trial enrichment strategies, and contribute to personalized prognostic modeling in Alzheimer's disease.

Ethical Integrity Statement

We confirm that this submission is our original work completed without the use of AI tools , that all sources are properly cited in APA 7 format, and that we understand BACSA is not responsible for any plagiarism or academic misconduct.

References

Albert, M. S., DeKosky, S. T., Dickson, D., Dubois, B., Feldman, H. H., Fox, N. C., Gamst, A., Holtzman, D. M., Jagust, W. J., Petersen, R. C., Snyder, P. J., Carrillo, M. C., Thies, B., & Phelps, C. H. (2011). The diagnosis of mild cognitive impairment due to alzheimer's disease: Recommendations from the National Institute on aging-alzheimer's association workgroups on diagnostic guidelines for alzheimer's disease. *Alzheimer's & Dementia*, *7*(3), 270–279. https://doi.org/10.1016/j.jalz.2011.03.008

Alzheimer's disease facts and figures. Alzheimer's Association. (2025). https://www.alz.org/alzheimers-dementia/facts-figures

Alzheimer's Disease Neuroimaging Initiative. (2023). ADNI data repository. https://adni.loni.usc.edu

Corder, E. H., Saunders, A. M., Strittmatter, W. J., Schmechel, D. E., Gaskell, P. C., Small, G. W., Roses, A. D., Haines, J. L., & Pericak-Vance, M. A. (1993). Gene dose of apolipoprotein e type 4 allele and the risk of alzheimer's disease in late onset families. *Science*, *261*(5123), 921–923. https://doi.org/10.1126/science.8346443

De Strooper, B., & Karran, E. (2016). The cellular phase of alzheimer's disease. *Cell*, *164*(4), 603–615. https://doi.org/10.1016/j.cell.2015.12.056

Hardy, J. (2006). Alzheimer's disease: The amyloid cascade hypothesis: An update and reappraisal. *Journal of Alzheimer's Disease*, *9*(s3), 151–153. https://doi.org/10.3233/jad-2006-9s317

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering, 9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55

Jack, C., Bennett, D., Blennow, K., Carrillo, M., Dunn, B., Haeberlein, S., Holtzman, D., Jagust, W., Jessen, F., Karlawish, J., Liu, E., Molinuevo, J., Montine, T., Phelps, C., Rankin, K., Rowe, C., Scheltens, P., Siemers, E., Snyder, H., … Silverberg, N. (2018). Nia-AA Research Framework: Toward a biological definition of

alzheimer's disease. *Alzheimer's & Dementia*, *14*(4), 535–562. https://doi.org/10.1016/j.jalz.2018.02.018

Mattsson, N., Andreasson, U., Zetterberg, H., & Blennow, K. (2017). Association of plasma neurofilament light with neurodegeneration in patients with alzheimer disease. *JAMA Neurology*, *74*(5), 557. https://doi.org/10.1001/jamaneurol.2016.6117

Moradi, E., Pepe, A., Gaser, C., Huttunen, H., & Tohka, J. (2015). Machine learning framework for early MRI-based alzheimer's conversion prediction in MCI subjects. *NeuroImage*, *104*, 398–412. https://doi.org/10.1016/j.neuroimage.2014.10.002

NumPy Developers. (2023). *NumPy (Version 1.x)* [Computer software]. https://numpy.org/

Palmqvist, S., Janelidze, S., Quiroz, Y. T., Zetterberg, H., Lopera, F., Stomrud, E., Su, Y., Chen, Y., Serrano, G. E., Leuzy, A., Mattsson-Carlgren, N., Strandberg, O., Smith, R., Villegas, A., Sepulveda-Falla, D., Chai, X., Proctor, N. K., Beach, T. G., Blennow, K., … Hansson, O. (2020). Discriminative accuracy of Plasma Phospho-Tau217 for alzheimer disease vs other neurodegenerative disorders. *JAMA*, *324*(8), 772. https://doi.org/10.1001/jama.2020.12134

Petersen, R. C. (2018). Practice guideline update summary: Mild cognitive impairment [retired]. *Neurology*, *90*(3), 126–135. https://doi.org/10.1212/wnl.0000000000004826

Project Jupyter. (2023). *Jupyter Notebook (Version 6.x)* [Computer software]. https://jupyter.org/

Python Software Foundation. (2023). *Python (Version 3.11)* [Computer software]. https://www.python.org/

Scikit-learn Developers. (2023). *Scikit-learn (Version 1.x)* [Computer software]. https://scikit-learn.org/

Shir, D., Graff-Radford, J., Hofrenning, E. I., Lesnick, T. G., Przybelski, S. A., Lowe, V. J., Knopman, D. S., Petersen, R. C., Jack, C. R., Vemuri, P., Algeciras-Schimnich, A., Campbell, M. R., Stricker, N. H., & Mielke, M. M. (2022, February 28). *Association of plasma glial fibrillary acidic protein (GFAP) with neuroimaging of alzheimer's disease and vascular pathology*. Alzheimer's & dementia (Amsterdam, Netherlands). https://pmc.ncbi.nlm.nih.gov/articles/PMC8883441/

Sperling, R. A., Aisen, P. S., Beckett, L. A., Bennett, D. A., Craft, S., Fagan, A. M., Iwatsubo, T., Jack, C. R., Kaye, J., Montine, T. J., Park, D. C., Reiman, E. M., Rowe, C. C., Siemers, E., Stern, Y., Yaffe, K., Carrillo, M. C., Thies, B., Morrison-Bogorad, M., … Phelps, C. H. (2011). Toward defining the preclinical stages of alzheimer's disease: Recommendations from the National Institute on aging-alzheimer's association workgroups on diagnostic guidelines for alzheimer's disease. *Alzheimer's & Dementia*, *7*(3), 280–292. https://doi.org/10.1016/j.jalz.2011.03.003

The pandas development team. (2023). *pandas (Version 2.x)* [Computer software]. https://pandas.pydata.org/